SIM Global Education

Professional Practice for Computer Scientists

CM2045

Part B -Sustainability In Computing

Lecture 6

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❑ Learning Objectives

- Identify the environmental impacts of computing systems, including energy consumption and e-waste.
- Explore strategies for designing sustainable software and hardware solutions that minimise environmental footprints.
- Evaluate sustainability and impact in the context of the computing industry.

❑ Recap on last lecture

- We have learnt 2 important approaches for sustainable computing:
  - Green software development
  - Green web design
- This week, we will focus on:
  - Renewable energy in tech
  - Green data centre
  - Circular IT and e-waste management
  - Decarbonized IT Operations

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Green data centres

❑ What is a data centre?

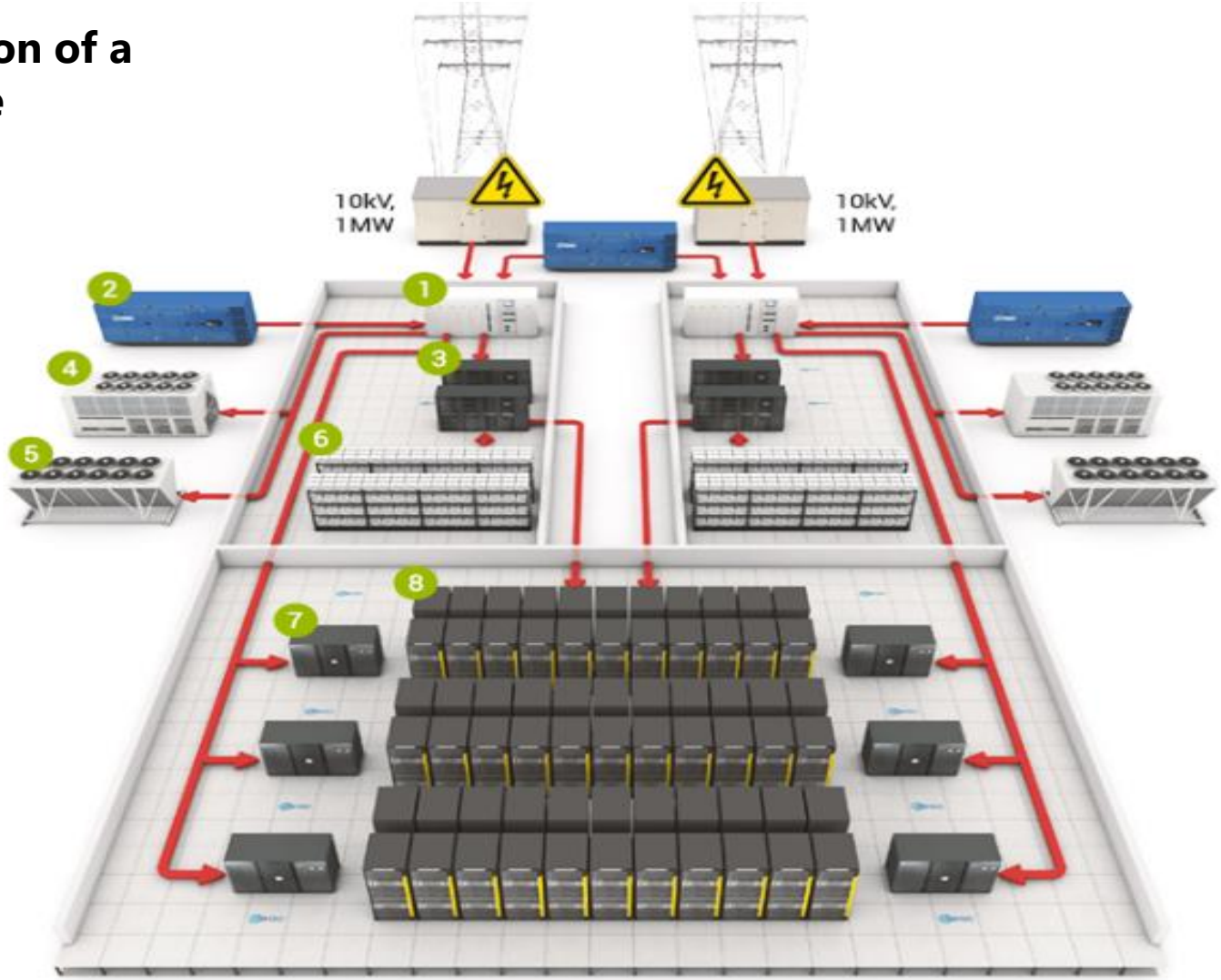
- A specially designed facility where computers, servers, and networking equipment are stored and operated. These powerful machines work together to manage and deliver digital services
- Data centres are the backbone of the internet.
- Size ranges from small 100ft2 cabinets up to massive 400,000ft2 “hyperscale” warehouses (Shehabi et al, 2016)
- Average data center demands around 5-10 megawatts (MW), but these large hyperscale facilities can be greater than 100 MW (IEA, 2025)

Green data centres

General classifications of data centres

- **By size and purpose:**
  - Enterprise Data Centers (private, on-premises)
  - Colocation Data Centers (leased space for third-party hardware)
  - Cloud Data Centers (hyperscale, e.g., AWS, Google Cloud)
  - Edge Data Centers (smaller, decentralized for low-latency processing)
  - Modular/Containerized Data Centers (portable, pre-fabricated units)

Visual representation of a general data centre



1 Main power distribution with Automatic Transfer Switch (ATS)
2 SDMO Diesel generator
3 APC Symmetra PX500 (Uninterruptible power supply)
4 Emerson Network Power chiller
5 DryCooler LU-VE
6 UPS batteries
7 Emerson Network Power climate control
8 Servers, storage, networking equipment

Source: https://blog.publiccomps.com/a-primer-on-data-centers/

Key Components of a data centre

- **Power Chain:** Utility → UPS → Generators → PDUs (Power Distribution Units)→ Racks.
- **Cooling Systems:** Chillers, CRACs (Computer Room Air Conditioners), airflow management.
- **Networking:** Switches, routers, cabling.
- **Redundancy:** Dual power paths, backup cooling.

Green data centres

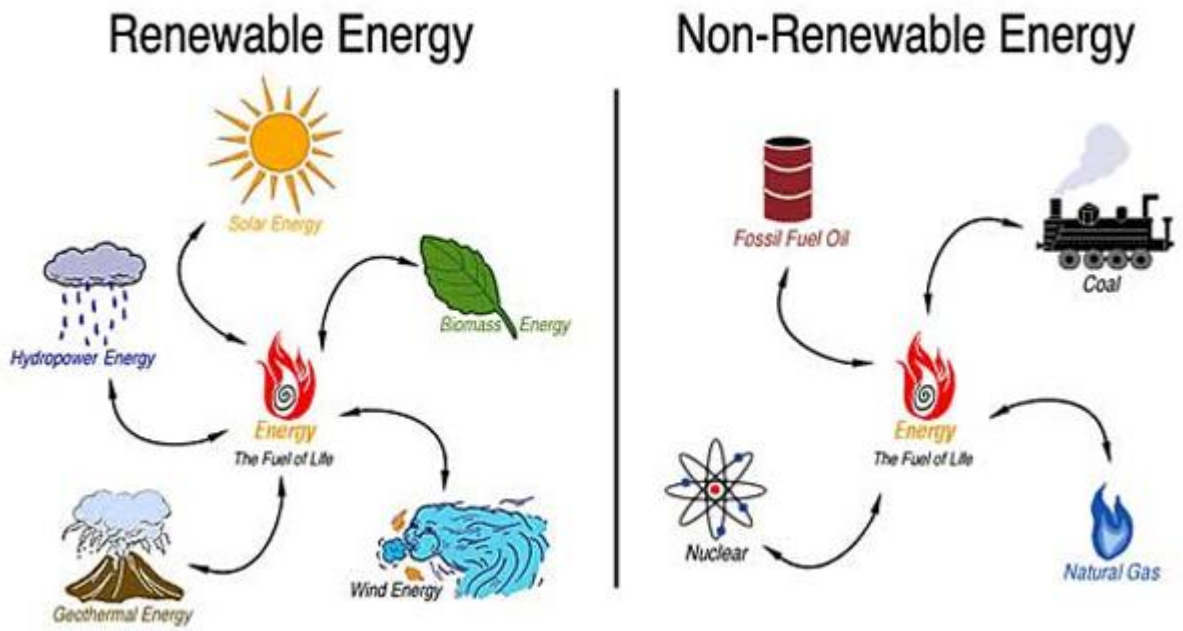
❑ Why do data centres consume a lot of energy?

- Always-On operation - Servers run 24/7, 365 days a year.
- Server emissions (CO_2) and other greenhouse gases released during the operation of data centres.
- Cooling alone can account for up to 40–50% of a data centre's total energy consumption.
- Data centres use:
  - Air conditioning
  - Liquid cooling systems
  - Fans and ventilation

Green data centres

❑ Energy sources for data centres

- Conventional (grid-powered): Sources include oil, gas and coal.
- Renewable-powered (solar, wind, hydro, geothermal and biomass)



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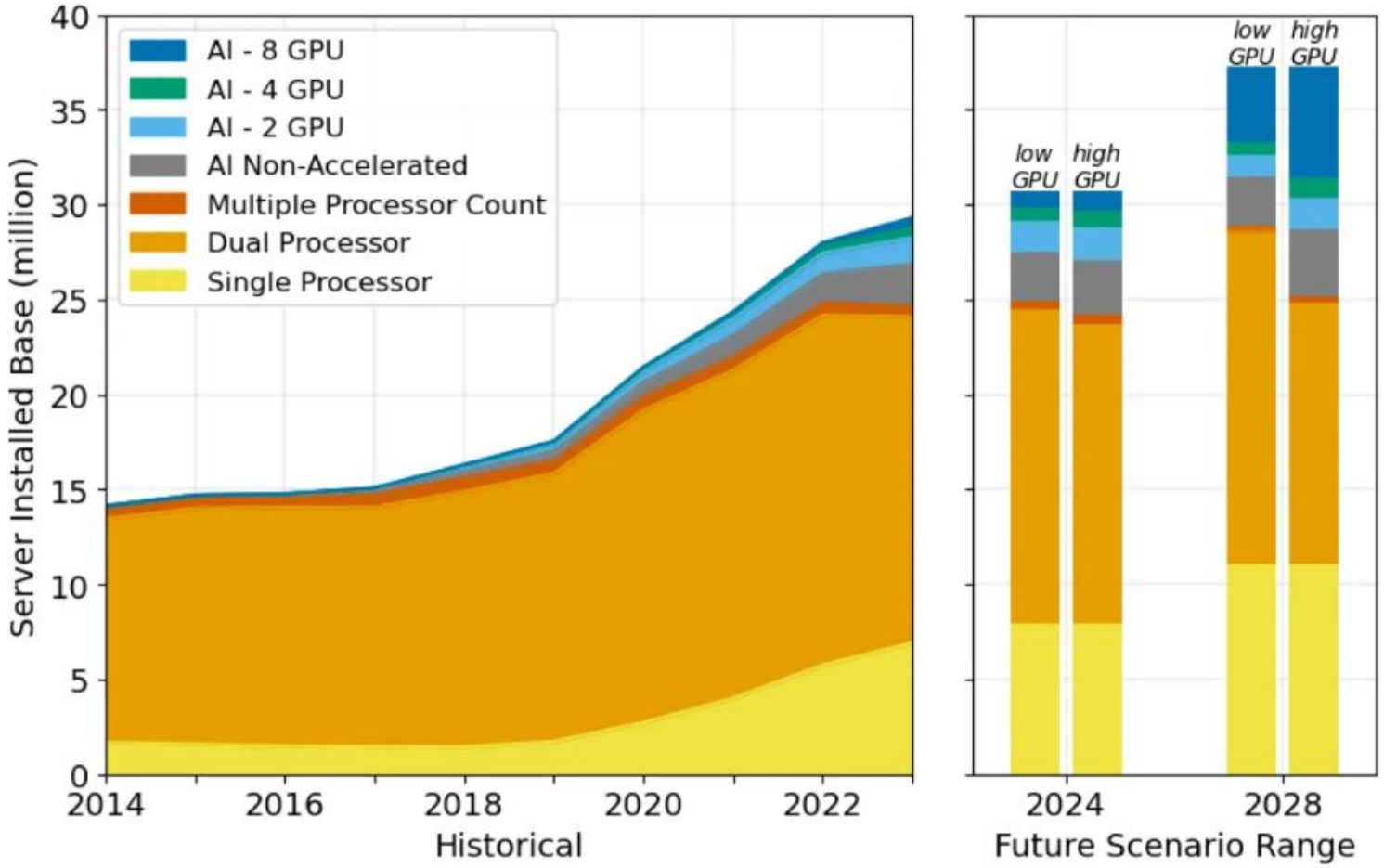
Green data centres

❑ Tremendous growth of servers installed from 2014 in US

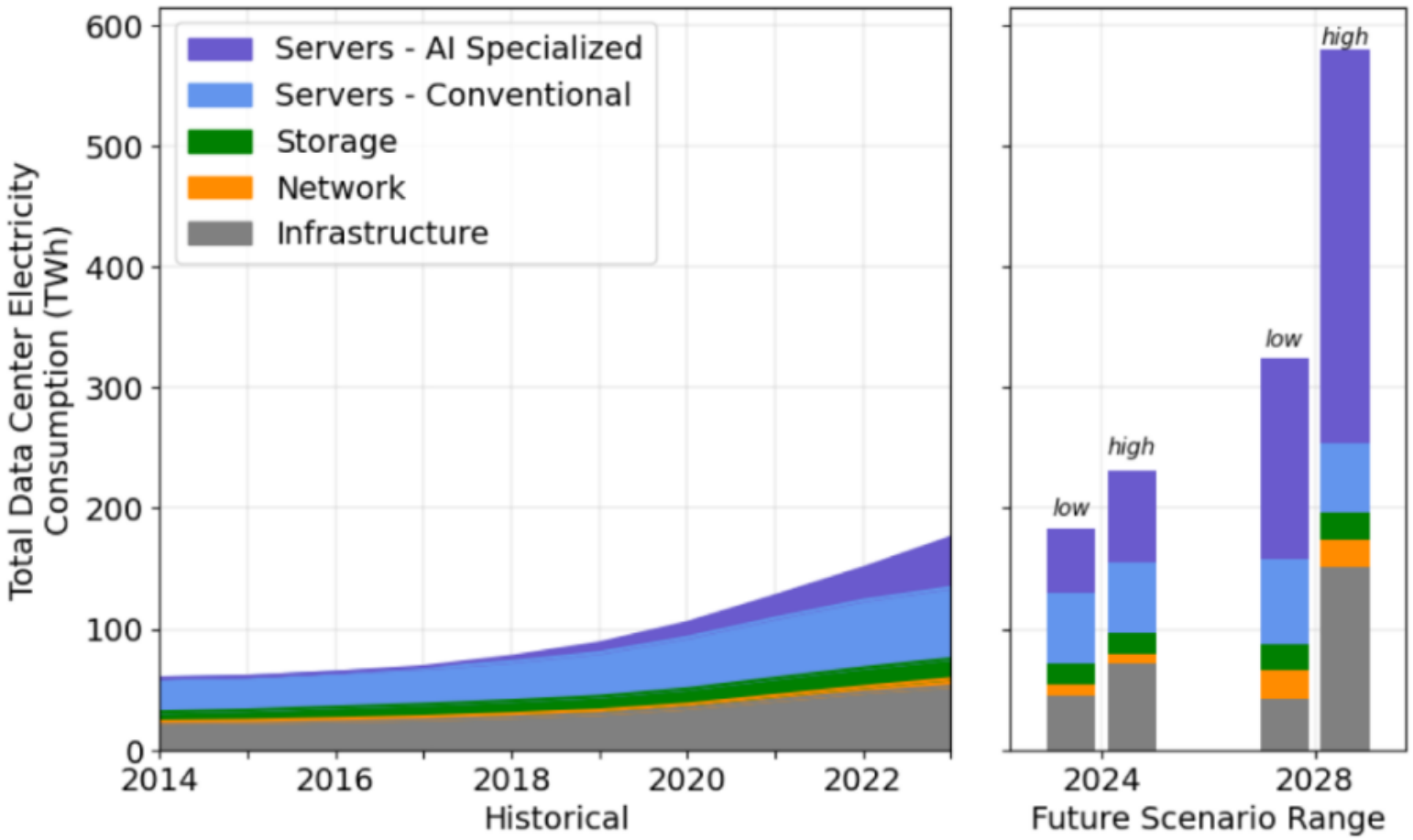


Source: Shehabi et al. 2024.

- 6.5 million servers were shipped in 2022.
- Estimated to rise to 7.7 million in 2028.
- This brings the total installed base from 14 million in 2014 to 21 million in 2020.
- Of those, 1.6 million are related to AI.

Green data centres

- **Data center energy consumption in the US from 2014**



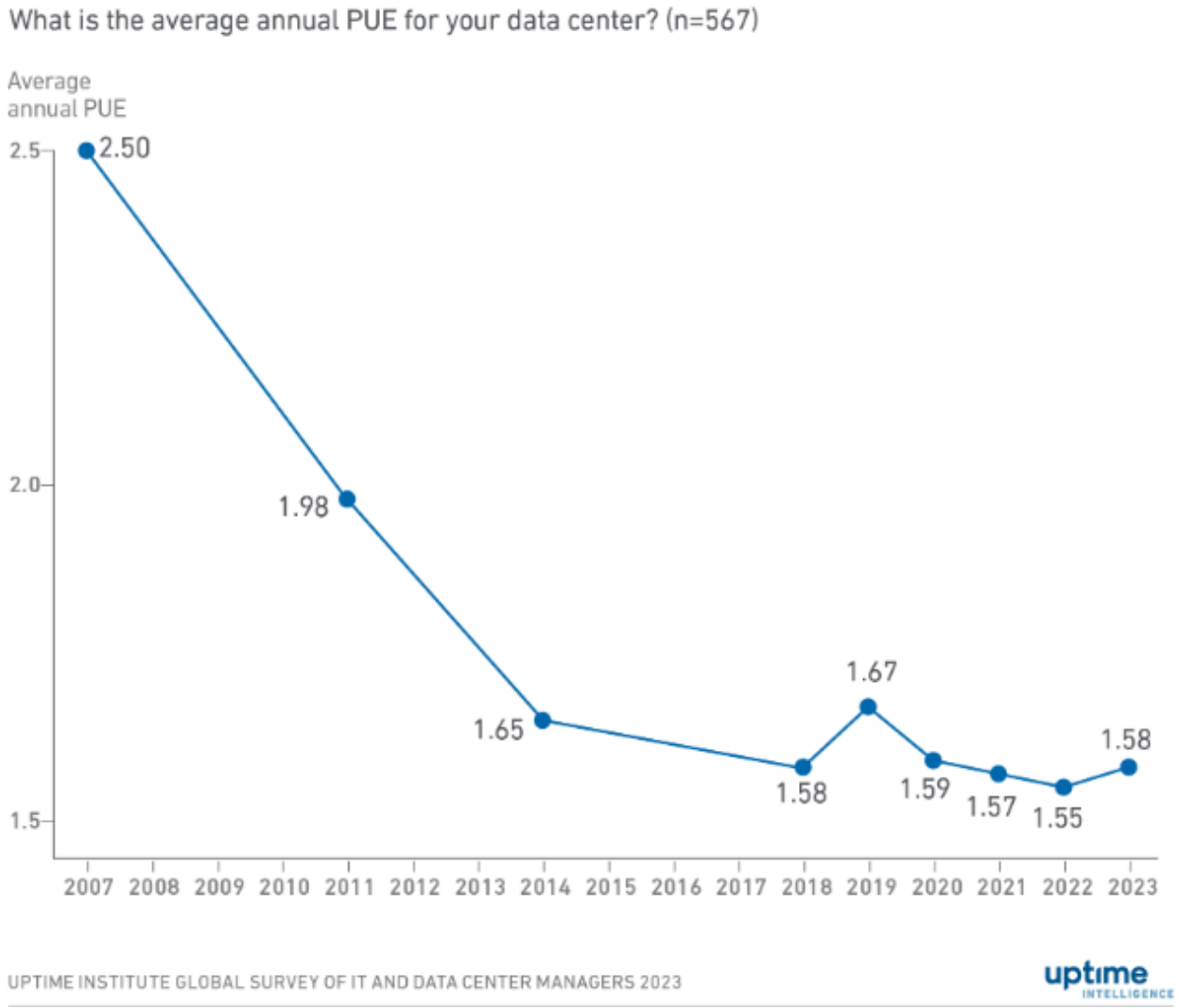
Source: Shehabi et al. 2024.

Green data centres

❑ How is energy efficiency measured?

- **PUE (Power Usage Effectiveness**) as a basic metric.
- PUE = Total Facility Energy ÷ IT Equipment Energy
- A simple ratio that compares the total energy used by the data centre to the energy used just for computing.
- E.g. A data centre uses 100 units of electricity, but only 60 units go to the servers, the rest powers things like cooling and lights. PUE = 100 ÷ 60 = 1.67
- What's a good PUE? The closer to 1.0, the better! 1.0 = Perfect Efficiency (all energy goes to servers, none wasted)
- Typical older data centres: PUE around 2.0. Modern efficient centres: 1.1 to 1.2. That's what companies like Google, Microsoft, and Meta aim for in their advanced data centres.

Green data centres



Key Findings:

- **Global PUEs have stalled**. Average PUE remains around 1.58, with little change in recent years.
- Despite better technology, **many data centres still struggle to improve efficiency**.
- **Cooling remains a major energy drain**—especially in older or less-optimized facilities. Advanced cooling techniques are not always adopted due to cost or complexity.
- **PUE varies by geography and climate**: Data centres in hotter climates tend to have higher PUEs due to extra cooling needs.

Source: https://journal.uptimeinstitute.com/global-pues-are-they-going-anywhere/

Types of data centre performance metrics

- Besides PUE. Other metrics include:
- **Thermal metrics** : Measures of cooling efficiency and temperature control.
  - Example: Cooling System Coefficient of Performance (COP), or Delta-T (temperature difference between supply and return air). Recommended server inlet temperatures between 18–27°C.
- **Greenhouse Gas (GHG) Emissions**: CO_2 or CO_2-equivalent emissions linked to energy consumption.
  - Example: kg CO_2 per kWh, or total tons of CO_2 emitted annually.
- **Humidity Metrics**: Control of humidity to improve cooling and reduce hardware failure risks
  - Example Metric: Relative Humidity (%RH) in data hall, ideally kept between 40–60%. Improper humidity control may cause static discharge (too dry) or condensation (too humid), impacting hardware lifespan.

Green data centres

❑ Drive for green data centres

- **Energy Consumption**: Data centers account for significant energy usage, with electricity costs being the largest operational expense.
- Example: U.S. data centers consume 61 billion KWh, approximately 1.5% of national electricity consumption (EPA report)
- **Growing pressure to reduce the environmental impact** of data centers, focusing on energy efficiency and sustainable operations.

Green data centres

❑ What makes a data centre “Green”?

- **Key characteristics of green data centres:**

➢ Efficient cooling systems

➢ Renewable energy sources

➢ Smart energy use through AI

➢ Efficient hardware and infrastructure

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Green data centres

❑ Key characteristics of a green data centre

- **Efficient cooling systems**: Reduce energy wasted on air conditioning.

➢ Examples:

- Free cooling: Uses outside air or water for cooling instead of traditional AC.
- Liquid cooling: Direct-to-chip liquid systems remove heat more effectively than air.
- Hot/cold aisle containment: Separates airflow paths to improve cooling efficiency.

- **Renewable energy sources**: Reduce greenhouse gas emissions by clean power.

➢ Examples:

- Solar panels installed on-site or nearby.
- Wind farms powering data centre clusters.
- Hydropower through green electricity providers.

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Key characteristics of a green data centre (cont'd)

- **Smart energy use through AI**: Maximise efficiency through automation and analytics.
  - Examples:
    - AI-based workload scheduling to shift processing to off-peak hours or greener times.
    - Dynamic cooling controls that adjust temperature and airflow based on real-time conditions.
    - Predictive maintenance to prevent energy waste from failing components.

- **Efficient hardware and infrastructure**: Reduce power consumption and e-waste.
  - Examples:
    - Low-power servers with high-performance-per-watt CPUs (i.e. eco-friendly processors).
    - Solid-state drives (SSDs) which consume less energy than traditional hard drives.
    - Modular, scalable infrastructure that prevents overprovisioning.

Green data centres

How big tech are addressing the rising energy demands of data centres?

https://www.youtube.com/watch?v=fZCaIaoDCgE

Green data centres

❑ Case Study: Google's Green Data Centres

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- **Uses sea water cooling system**
  - Replaced traditional energy hungry cooling system by high-tech seawater cooling system.
  - Achieves an excellent Power Usage Effectiveness (PUE) of 1.1.
- **100% renewable energy matching**: Google purchases enough renewable energy (like wind and solar) to equal all the electricity it uses worldwide each year."
- **Carbon-free goal (24/7 by 2030):** Plan to use clean energy every hour of every day, by 2030. This means their servers will run on green power 24/7.
- **Geothermal (electricity from underground heat) expansion**: Scaling from 3.5 MW to 115 MW over six years.
- **AI optimization**: DeepMind AI reduced cooling energy use by ~40%.

Green data centres

❑ Case Study: Microsoft's Green Data Centres

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- **Project Natick – Underwater data centre**:
  - Placed a data center on the seafloor near Scotland. The cold water naturally cooled the equipment, resulting in a PUE of 1.07, and the project operated fully on renewable power.
- **Water-efficient cooling systems**:
  - Many of Microsoft's data centers use cooling techniques that minimize water use—important in areas with droughts.
- **LEED Gold certification program:**
  - Committed to building all its data centers to LEED Gold standards, which are guidelines for environmentally friendly buildings. This includes energy savings, water efficiency, and sustainable materials.

Renewable energy in tech

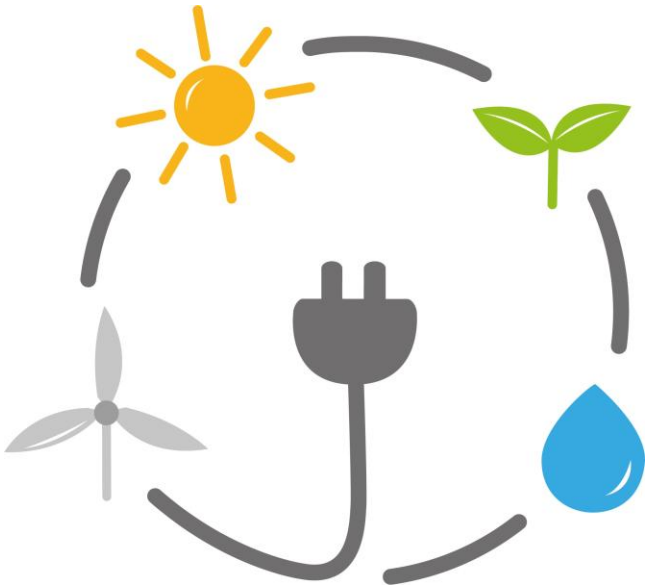
Renewable energy 101

https://www.youtube.com/watch?v=1kUE0BZtTRc

Renewable energy in tech

❑ Types of renewable energy

- **Solar energy**: Harnesses sunlight using photovoltaic panels or solar thermal systems.
- **Wind energy**: Converts wind flow into electricity using turbines.
- **Hydropower**: Generates electricity by using water flow, typically from dams.
- **Geothermal energy**: Utilizes heat from beneath the Earth's surface for electricity generation and direct heating.
- **Biomass energy**: Involves burning organic materials like wood, agricultural residues, or waste to produce energy.



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Renewable energy in tech

ADVANTAGES	DISADVANTAGES
✓ Renewable energy will not run out.	❖ Renewable energy has high upfront costs.
✓ Renewable energy has lower maintenance requirements.	❖ Renewable energy is intermittent.
✓ Renewables save money.	❖ Renewables have limited storage capabilities.
✓ Renewable energy has numerous environmental benefits.	❖ Renewable energy sources have geographic limitations.
✓ Renewables lower reliance on foreign energy sources.	❖ Renewables aren't always 100% carbon-free.
✓ Renewable energy leads to cleaner water and air.	
✓ Renewable energy creates jobs.	
✓ Renewable energy can cut down on waste.	

Source: https://www.energysage.com/about-clean-energy/advantages-and-disadvantages-of-renewable-energy/

Renewable energy in tech

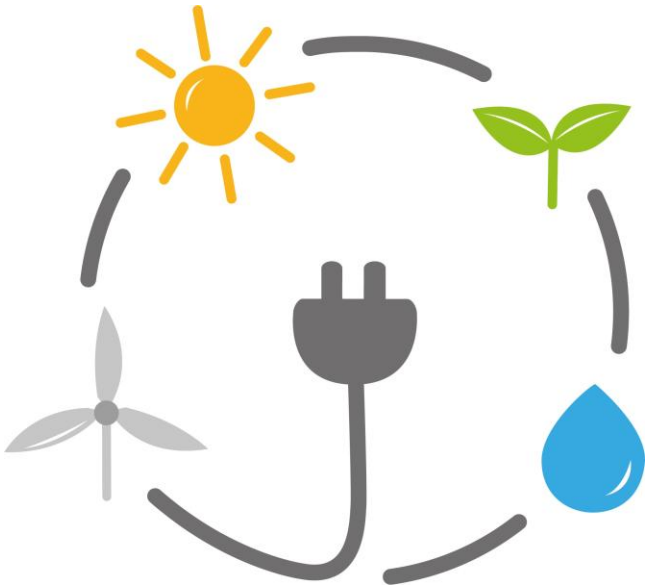
❑ Reasons why renewable energy is not always 100% clean?

- **Manufacturing & installation impacts :**
  - Solar panels, wind turbines, and batteries require mining of rare-earth metals (like lithium, cobalt, and neodymium).This mining can cause environmental degradation (e.g., deforestation, water pollution).
- **Land & habitat disruption:**
  - Large-scale solar farms, wind farms, and hydroelectric dams can disturb ecosystems, lead to habitat loss for animals, alter natural waterways (especially with dams).
- **Hydropower's ecological Impact:**
  - Dams can disrupt fish migration, alter river ecosystems and lead to methane emissions from submerged vegetation in tropical areas.

Renewable energy in tech

❑ Global Renewable Energy Outlook (2024–2030)

- Renewable energy use is projected to grow by 60% by 2030. (power, heat, transport)
- Renewables' share in energy use to rise from 13% (2023) to ~20% by 2030.
- At COP28, countries agreed to **triple** global renewable energy capacity **by 2030**. Recent COP30 (in Nov 2025) hosted by Brazil ended with controversial outcomes and disappointments.
- Global renewable energy capacity is projected to increase by 2.7 times by 2030, which is 25% higher than what countries have currently planned, but still falls short of the goal to triple capacity.
- Sustainable practices, responsible sourcing, and technology improvements are essential to truly make renewable energy environmentally friendly



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E-Waste management

https://www.youtube.com/watch?v=HV2Ir1e513o

E-Waste management

❑ Why e-Waste matters?

- Electronic waste (e-waste) includes discarded phones, laptops, TVs, etc.
- Fastest-growing solid waste streams globally, with an annual increase of 3%–5%.
- E-waste increased 53.6 million metric tonnes in 2019 globally. Only 17.4% was formally collected and recycled. (Forti et al., 2020)
- Toxic components (e.g. lead, mercury, cadmium) contaminate soil, air, and water => poses significant environmental and health risks.
- Illegal trade and informal recycling practices increases risks.
- E-waste dumping in low-income / developing countries exploits lax regulations.

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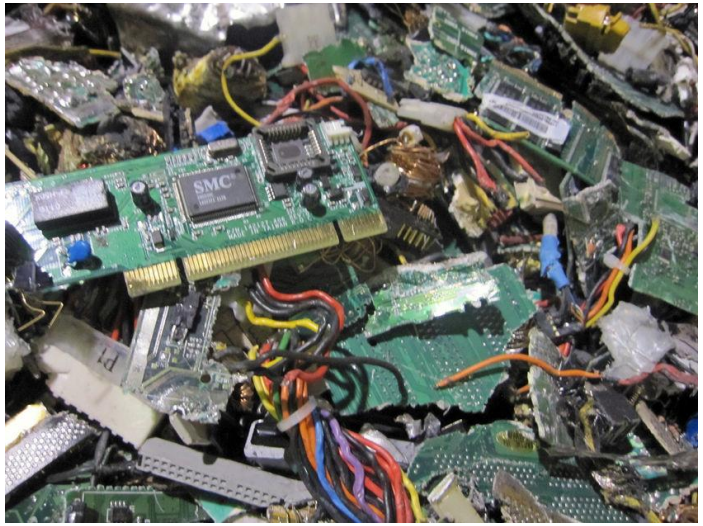
E-Waste management

https://www.youtube.com/watch?v=OTqgYvNSA4c

E-Waste management

❑ Why e-Waste matters?



- **Real-life examples:**

- **Toxic E-Waste Dumped in Poor Nations, Says United Nations** https://ourworld.unu.edu/en/toxic-e-waste-dumped-in-poor-nations-says-united-nations

- **Electronic Waste in Guiyu, China: A City under Change?** https://www.environmentandsociety.org/arcadia/electronic-waste-guiyu-city-under-change

- **Malaysia -Tighten controls on waste processing, ban importation of e-waste** https://www.nst.com.my/opinion/letters/2025/03/1188600/tighten-controls-waste-processing-ban-importation-e-waste?utm_source=nst&utm_medium=relatedarticles

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E-Waste management

❑ Global e-waste landscape

- **Transboundary movement:**

➢ Despite the Basel Convention (1992) banning the export of hazardous waste to developing countries, illegal shipments of e-waste **continue, especially to West Africa and Southeast Asia.**

- **Informal recycling practices:**

➢ In countries like Ghana, India, and Nigeria, e-waste is often processed by burning, acid leaching, or manual dismantling with no protective gear. Results in air, water, and soil contamination.

- **Environmental impact**: Heavy metals cause persistent organic pollutants to the environment. .

➢ Source: Forti et al., The Global E-waste Monitor 2020; Puckett et al., "Exporting Harm," Basel Action Network.

E-Waste management

❑ Strategies for sustainable e-waste management

- **Legislative management**
  - Enforce Extended Producer Responsibility (EPR): Require manufacturers to take back and responsibly recycle their products.
  - Ban illegal imports/exports of e-waste: Strengthen customs checks and cross-border cooperation under the Basel Convention.
  - Implement "Right to Repair" laws: Ensure consumers can repair devices easily, extending their lifespan.
    - Example: The EU's Waste Electrical and Electronic Equipment (WEEE) Directive mandates collection targets and producer accountability.

E-Waste management

❑ Strategies for sustainable e-waste management

- **E-waste recycling practices:**
  - ➢ Adopt clean and closed-loop recycling technologies: Reduce emissions and safely recover precious metals and rare earth elements.
  - ➢ Support local e-waste recycling infrastructure: Especially important in developing countries where informal processing dominates.
    - Example: Singapore's ALBA E-waste Smart Recycling Programme includes high-tech sorting and material recovery systems.

E-Waste management

❑ Strategies for sustainable e-waste management

- **Eco-friendly product design**
  - ➢ Design for disassembly: Use fewer screws, glue-free parts, and modular design to ease recycling.
  - ➢ Replace toxic components: Eliminate PVC, BFRs, and heavy metals from electronics.
  - • Example: Fairphone designs smartphones with replaceable modules, prioritizing ethical sourcing and recyclability.
  - • Sources: Greenpeace Guide to Greener Electronics, 2017

E-Waste management

❑ Strategies for sustainable e-waste management

- **Public awareness & engagement**
  - ➢ Run national campaigns: Promote recycling drop-offs, repair culture, and responsible digital consumption.
  - ➢ Label electronics clearly: Inform consumers about hazardous components and recycling guidelines.
  - Example: The EPA's (US Environmental Protection Agency) Sustainable Materials Management program encourages schools, communities, and businesses to host collection events.
  - Sources: EPA, 2024; WebFX, 2023

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❑ What can IT professionals do to combat e-waste?

- **Personal Responsibility (as Individuals)**
  - Practice repair culture: Extend the life of devices through regular maintenance and upgrades.
  - Recycle responsibly: Use certified e-waste recycling centers (e.g., e-Stewards or R2 certified).
  - Donate or resell: Give functional equipment a second life through charities, schools, or resale platforms.
  - Minimize tech consumption: Avoid unnecessary device upgrades; choose modular, repairable devices.

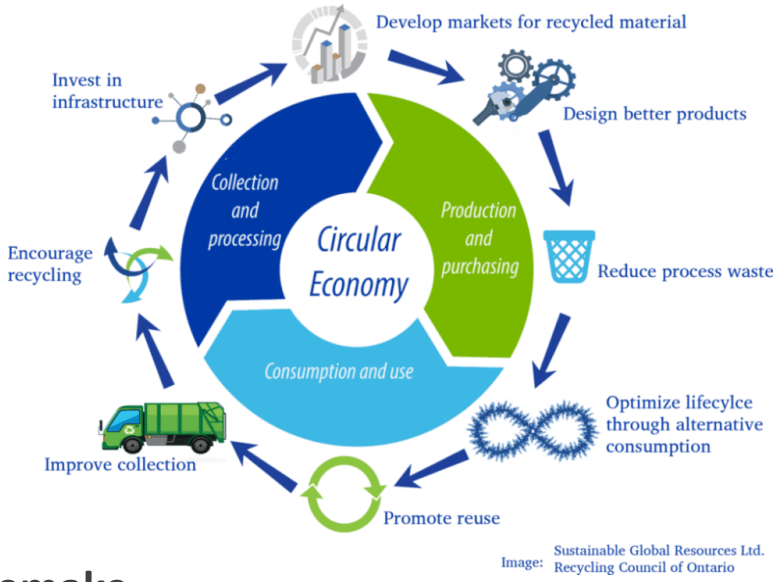
❑ What can IT professionals do to combat e-waste?

- **Corporate Responsibility (as IT Decision-Makers)**
  - Implement IT asset disposition (ITAD) policies: Ensure proper data wiping, secure disposal, and environmentally responsible recycling.
  - Green procurement: Choose vendors with EPEAT or ENERGY STAR certifications and transparent take-back programs.
  - Reduce hardware footprint: Use virtualization and cloud services to reduce dependency on physical hardware.
  - Engage in **Circular Economy**: Participate in device leasing, refurbishing, or buy-back programs.
  - E.g: Google & Dell run take-back programs and design hardware with recyclability.

❑ Introduction to Circular Economy

- Circular economy regenerates nature by maintaining, reusing, recycling materials in use, and composting products. It aims at reducing waste, pollution, and reliance on finite resources while addressing climate change and biodiversity loss.

- Three key principles:
  - ➢ Eliminates waste and pollution
  - ➢ Circulate products and materials
  - ➢ Regenerates nature



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- **Linear econonmy (traditional)** = **Take → Make → Use → Dispose**

➢ **Circular Economy** = **Make → Use → Reuse → Repair → Recycle → Remake**

- Watch this video: https://www.youtube.com/watch?v=NBEvJwTxs4w

- More reading at: https://www.ellenmacarthurfoundation.org/topics/circular-economy-introduction/overview

Sustainability compliance in computing

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❑ Key International Standards on Sustainability

❑ **Paris Agreement (2015):**

- A global accord within the United Nations Framework Convention on Climate Change (UNFCCC) aimed at limiting global warming to below 2°C.
- Encourages tech companies to reduce carbon emissions, promote renewable energy use, and invest in low-carbon technologies.
- **The European Union's Green Deal:**

➢ The European Commission's European Green Deal sets ambitious targets for carbon neutrality by 2050, impacting all industries, including IT.

➢ Encourages renewable energy use in data centers and mandates sustainability reporting for tech companies.

Sustainability compliance in computing

❑ Key International Standards on Sustainability

- **ISO 14001:2015 (Environmental Management Systems):**

➢ Provides a framework for environmental management that can be applied to data centers and IT infrastructure.

➢ Encourages organizations to assess environmental risks, reduce waste, and enhance resource efficiency in IT operations.

Key International Standards on Sustainability

- **EU's WEEE (Waste Electrical and Electronic Equipment) directive: Managing e-Waste in Europe**
  - An EU legislative framework first adopted in 2003, revised in 2012 (Directive 2012/19/EU).
  - Reduce e-waste generation and improve collection and recycling rates.
  - Recover valuable materials and reduce landfill.
  - Promote eco-design to make products easier to reuse, repair, and recycle.
  - Ensure producer responsibility: manufacturers must finance take-back and recycling systems.

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❑ Strategies for sustainable computing for companies

- Ensure energy reduction for data centres
- Adopt Circular IT and the green computing mindset
- Extending hardware longevity to cut e-Waste
- Adopt Device Lifecycle Management (DLM)
- Implement repair-first policies
- Encourage internal reuse
- Work with ITAD (IT Asset Disposition) vendors to ensure responsible reuse and recycling
- Decarbonized IT operations

Emerging trends of sustainability in computing

❑ Key emerging trends

- Rise of renewable energy in data centers
- Energy-efficient hardware and software
- Circular economy and e-Waste management
- Carbon neutrality and emission reduction
- Green software engineering practices
- Use of AI for sustainability in computing

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Emerging trends of sustainability in computing

❑ AI for Sustainability in Computing

- **AI in Energy Management**: AI algorithms can optimize energy use in data centers and reduce power consumption.
- **Predictive Maintenance**: AI-driven systems predict hardware failures, reducing waste by extending the life of equipment.
- **Examples**:

➢ **Google DeepMind**: Uses AI to predict and optimize cooling systems in data centers, saving energy.

➢ **AI in Smart Grids**: AI can optimize power distribution in smart grids, ensuring more efficient use of renewable energy.

ACTIVITY

Quick Sustainability Evaluation

- Time (25 mins)
- Evaluate the sustainability policies and practices of the National University of Singapore (NUS).
- Go to https://docs.google.com/document/d/1woLndR45HWKYrtvyr0agnrSjDTnJGar2aAYKRGBaTEQ/edit?usp=sharing
- Work in pairs or group. Type your answers.
- Class discussion.

Lecture 6 - recap

ACTIVITY

What have you learnt today?

- Go to **joinmyquiz.com** to take a quiz.

Lecture 6 recap

You should be able to answer these key questions:

1. How do green data centres reduce energy consumption and carbon emissions?
2. What is Power Usage Effectiveness (PUE), and why is it important?
3. What role does renewable energy play in sustainable computing?
4. Why is e-waste a growing global concern, and what are its environmental impacts?
5. What are some effective strategies for managing e-waste sustainably?
6. What makes a product or hardware design eco-friendly?
7. How do international regulations like the Paris Agreement and WEEE Directive support green computing?
8. What is the concept of Circular IT, and how does it support sustainability?
9. How is AI being used to improve energy efficiency and sustainability in IT operations?**

Reference Lecture 6

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